

An introduction to Reinforcement Learning

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Reinforcement Learning?

- Agenda:
 - What is Reinforcement Learning
 - Key terms in RL
 - A basic RL Problem
 - Markov Decision Process
- QnA

The essence of Reinforcement Learning

- Algorithms learn by trial and error, like how a person might learn to play chess.
- Process of RL: Machines interact with their environment, receive feedback, and refine their actions to improve performance.

Key Terms

- **Agent:**
 - ❖ The entity that learns by interacting with the environment. Examples are, a robot learning to walk or AI learning to drive.
- **State:** The current situation or position of the agent in the environment. Note: Changes as the agent moves.
- **Environment:** The world or context within which the agent operates. Note: The state is derived from the environment.

Key Terms - Continued

- **Action:** The decision or step the agent takes after observing the environment.
- **Reward:** The feedback the agent receives after an action. Note: Positive rewards for correct actions, negative rewards for incorrect actions.
- **Policy:** The strategy the agent uses to decide actions in particular states.
- **Goal:** The objective the agent aims to achieve through learning.

Recognizing a RL problem

- If we can identify the following things, we can visualize and solve the task as a RL problem:
 - ❖ Agent: Entity performing actions to achieve a goal.
 - ❖ Environment: The context or world within which the agent operates.
 - ❖ Action: The movement that agent can make within the environment taking it closer to or farther from the goal
 - ❖ Goal: Desired state of the agent

Types of Predictions

- **Model-Free Prediction:**

- ❖ Definition: Agent learns without knowing the environment.
- ❖ Example: Exploring a new city without a map.

- **Model-Based Prediction:**

- ❖ Definition: Agent learns with full knowledge of the environment.
- ❖ Example: Navigating a familiar house.

Understanding Rewards

- Immediate Rewards: Feedback given right after an action.
- Delayed Rewards: Feedback given after a set of actions.
- Episodes: A series of actions and states leading to a reward.
- Example Scenario, a race competition:
 - ❖ Mission: Run 100 meters in the shortest time.
 - ❖ Intervals: Reward given every 10 meters.
 - ❖ Learning: Adjust actions (e.g., leaning forward) for better results.

Markov Decision Process (MDP)

- **Definition:** A framework for modeling decision-making where outcomes are partly random and partly under the control of the decision-maker.
- **Components:**
 - ❖ States (S): Different situations the agent can be in.
 - ❖ Actions (A): Possible decisions the agent can take.
 - ❖ Transition Probabilities (P): Probability of moving from one state to another after an action.
 - ❖ Rewards (R): Feedback received after transitioning from one state to another due to an action.
 - ❖ Policy (π): Strategy that defines the action the agent should take in each state.

MDP Example

- **Navigating a Grid**
 - ❖ States: Positions on the grid.
 - ❖ Actions: Move up, down, left, right.
 - ❖ Transition Probabilities: Chance of moving to the next position based on the action.
 - ❖ Rewards: Points for reaching certain positions or penalties for hitting obstacles.
 - ❖ Policy: Plan to maximize total reward by navigating the grid efficiently.
- **Markov Property:** Future state depends only on the current state and action, not on the sequence of events that preceded it.
- **Goal:** Find a policy that maximizes cumulative rewards over time.

Solving a RL problem

- **Bellman Equation:** Used to solve the Markov Decision Process (MDP).
- **State Value Function ($V(s)$):**
 - Definition: Expected total reward from a particular state to the goal.
 - Example: Navigating to collect fruits in a room.
 - Calculation: Sum of rewards from the current state to the goal state.
 - Note: Different policies lead to different state values.

Solving a RL problem - Continued

- **Action Value Function ($Q(s, a)$):**
 - ❖ Definition: Expected total reward considering both, state and the next action.
- **Optimal Policy:**
 - ❖ Goal: Identify the best path that maximizes total rewards.
 - ❖ Method: Use the action value function to determine the optimal actions to take in each state.

Getting everything so far?

Let's find out

Quiz

Question 1: What is the position of an agent at a given time?

- A. state
- B. location
- C. scene
- D. environment

Quiz

Question 2: How is a model-free prediction different from a model-based prediction?

- A. Model-based prediction is a prediction which is based on a model
- B. For model-free prediction, the agent has no idea of what the environment looks like while for model-based prediction, the agent has full knowledge of the environment.
- C. Model-free prediction is the prediction of a free model.
- D. For model-free prediction, the agent has full knowledge of the environment while for model-based prediction, the agent has no idea of what the environment looks like.

Quiz

Question 3: What is the Markov decision process used for?

- A. It is used to solve a reinforcement learning problem.
- B. It is used to design decision processes.
- C. It is used to make Markov decisions.
- D. It is used to represent a reinforcement learning problem mathematically.

Quiz

Question 4: What is a state-value function?

- A. It is the expected value of the reward to the goal from a particular state s .
- B. It is the actual value of the reward gotten by the agent after taking an action in a state.
- C. It is used to mark a state.
- D. It is the value of a state.

Quiz

Question 5: How is an action-value function different from a state-value function?

- A. The state value function stores the states while the action value function stores the actions.
- B. They both store the same value except that the action value is recorded after an agent takes an action while state-value is the reward in a current state.
- C. They are not different.
- D. The state-value function is the expected reward of an agent on reaching the goal starting at state s while action-value function is the expected reward of the agent starting at state s and taking an action a .